

Technology Stack

5. Technology Stack and Justification

- **Python** — readable, strong AI ecosystem, quick to learn.
- **FastAPI** — modern, fast web framework with built-in input validation and automatic API documentation.
- **Uvicorn** — server used to run FastAPI.
- **Jinja2 templates with HTML/CSS/JavaScript** — simple page rendering without a heavy front-end framework.
- **Google Gemini (google-generativeai)** — provides text and image understanding for recommendations.
- **python-jose** — creates and verifies JWT tokens.
- **Passlib with bcrypt** — safe password hashing.
- **Pillow** — opens the uploaded outfit image before sending it to Gemini.
- **python-multipart** — lets FastAPI read form and file uploads.
- **python-dotenv** — loads settings from a .env file.